

Methunraj A

Bachelor of Technology in Computer Science and Engineering
SRM Institute of Science and Technology, Tiruchirappalli

+91-7708847437

methunraja7393@gmail.com

LinkedIn | GitHub

EDUCATION

- Bachelor of Technology – Computer Science and Engineering (Honors: Quantum Computing)** 2023 – 2027
SRM Institute of Science and Technology, Tiruchirappalli CGPA: 9.50/10

TECHNICAL SKILLS

Programming & Databases: Python, C/C++ (Object-Oriented Programming), SQL
Data Science & ML: Pandas, NumPy, Scikit-learn, PyTorch, XGBoost, Statistical Modeling
NLP & Foundation Models: spaCy, Sentence-BERT, NER, Image Processing, Object Recognition
Data Visualization : Matplotlib, Seaborn, Plotly, Dashboard Development
Tools & Platforms: Git, Jupyter, Snowflake, FastAPI, REST APIs

PROFESSIONAL EXPERIENCE

- Data Science Intern – AI & NLP** Aug 2025 – Oct 2025
Infosys Springboard Remote
 - Engineered an end-to-end NLP pipeline for automated skill extraction, utilizing spaCy for Named Entity Recognition (NER) and custom Regex to process unstructured PDF and DOCX documents
 - Implemented Transformer-based Foundation Models (Sentence-BERT) to generate vector embeddings, computing a Cosine Similarity Matrix in NumPy to dynamically score semantic resume-job matches
 - Developed and deployed a full-stack interactive web application using Streamlit, integrating the machine learning backend with Plotly for dynamic, real-time data visualization of candidate skill gaps
- Research Intern – Computer-Aided Manufacturing** Jun 2025 – Jul 2025
Defence Research & Development Organisation (DRDO), CVRDE, Avadi Onsite
 - Developed optimized CNC toolpath algorithms using quintic Bézier curves with Python, performing control point extraction and Bernstein polynomial analysis for precision manufacturing
 - Created data visualization dashboards for process monitoring and analysis using Matplotlib and NumPy

PROJECTS

- Real-Time Air Quality Prediction System** *Python, Scikit-learn, XGBoost, Flask, REST APIs*
 - Engineered a high-performance Stacking Ensemble model (Random Forest, XGBoost, LightGBM, CatBoost with a Ridge Meta-Learner) to predict hyper-local PM2.5 levels using advanced spatiotemporal feature engineering
 - Developed a Flask REST API that dynamically orchestrates multiple live external services (WAQI, OpenWeatherMap, OpenRouteService) to calculate and return the "Fastest and Cleanest" geographical routing paths
 - Implemented production-ready backend architecture utilizing Flask-Limiter for network rate limiting and efficient in-memory data structures (deque) to cache time-series station history
- The Last Spark– iOS Arcade Game (Apple Swift Student Challenge Winner)** *SwiftUI, AVFoundation*
 - Engineered a globally recognized iOS game demonstrating advanced SwiftUI architecture; built a custom game loop and dynamic difficulty engine optimized for high frame rates.
 - Integrated Apple frameworks (CoreHaptics, AVFoundation) to create an immersive user experience, achieving seamless animations and fluid state management.
- Lost and Found Management System** *React, Flask, MySQL, Python, Scikit-learn*
 - Built full-stack web application using OpenAI's CLIP (Foundation Model) for advanced semantic text-to-image matching; implemented SQL database design with complex queries
 - Developed Flask REST API backend with authentication, data validation, and automated image processing pipeline

AWARDS & CERTIFICATIONS

- Apple Swift Student Challenge 2026 Winner** – Selected among thousands globally for developing interactive iOS app demonstrating innovative use of SwiftUI and Apple frameworks
- SnowPro Associate: Platform Certification** – Snowflake
- NPTEL Database Management System** – SQL, NoSQL, and database design